

# Kimchhorn Sambath

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## Skills

**Languages:** Java, Python, JavaScript, TypeScript, SQL, HTML, CSS, Rust, Swift

**Technologies:** React, Next.js, Node.js, REST APIs, PostgreSQL, Git, GitHub, Docker, AWS, Jest, Playwright, Postman

**Practices:** SDLC, CI/CD, unit testing, integration testing, code review, debugging, automation, version control

**Methodologies:** Agile (Scrum/Kanban), TDD, Unit, Integration, End-to-End, A/B, and White/Black Box Testing

**Developer Tools:** Visual Studio Code, UML, Jira, Notion, Figma, Vercel, Azure OpenAI, Gemini API

**Certifications:** “The Complete Full-Stack Web Development Bootcamp (Udemy)” - HTML, Node.js, SQL

“JPMorgan Chase Software Engineering Virtual Experience (Forage)” - Java, Maven, Kafka

## Work Experience

**Research Assistant** | *Simon Fraser University*

May 2025 – Present

- Collaborate with Prof. **Saba Alimadadi** [↗](#) in an Agile research workflow to improve the testability of **JavaScript** and **TypeScript** systems through **static analysis** and **automated test generation**.
- Developed **TypeScript AST-based analysis** and **validation tooling** to detect anti-patterns and testability issues in source code, including patterns that hinder test generation, weaken observability and add complexity.
- Designed a research pipeline that identifies key code components used by the system’s **LLM-based** test generation workflow, achieving **98%** accuracy against corrected code and **benchmark** suites developed by collaborating teams.
- Developed **Jest-based validation tests** for the AST analysis system, reviewing failures weekly with the team to uncover edge cases and improve framework reliability.
- Authored onboarding **documentation** for incoming teammates and contributed to a research paper to communicate project findings clearly and support **future publication**.

## Projects

**Ceasefire.com** | *PostgreSQL, Python, Next.js, React.js, TypeScript, Docker, AWS, Kubernetes*

Feb 2026 – Present

- Design and develop a full-stack platform to share humanitarian stories on conflict zones and refugee experiences.
- Build automated web scraping pipelines using **Docker**, **Kubernetes**, and **AWS** for scalable, reliable global access
- Build GitHub Actions workflows to scrape content **every 12 hours** using Selenium and Python, then normalize and load the data into Supabase Postgres.

**KeyGod** [↗](#) | *Rust, Swift, WebSocket, Enigo (OS-level input simulator), Tokio*

Oct – December 2025

- Built a **Swift iOS** client and **Rust desktop** bridge to send on-screen and **hardware keyboard** input from an **iPad to a PC**, with normalized key and modifier mappings for cross-platform OS-level input injection.
- Designed a typed **WebSocket protocol** for text, key, ping, and authenticated messages, with heartbeat handling, malformed-frame rejection, and secure command paths to improve communication reliability.

**Drip.ai (StormHacks) (Live Demo)** [↗](#) | *Next.js, TypeScript, React, Tailwind CSS, Gemini API, Vercel*

Oct 2025

- Led a 4-person team on an **AI-powered wardrobe**, integrating **Virtual try-on** and a **Gemini AI stylist**.
- Integrated React frontend features with backend **REST API** endpoints, handling asynchronous requests, JSON responses, and client-side updates for styling workflows.

**Kim’s Portfolio 2.0 (Live Demo)** [↗](#) | *Next.js, React, TypeScript, Tailwind CSS, Anthropic API*

Aug – Sept 2025

- Modernized a static **HTML/CSS/JavaScript portfolio** [↗](#) in **Next.js**, introducing reusable components.
- Built a **Claude-powered chatbot** that lets visitors ask about **projects**, **technical skills**, and **experience** directly.

**Deep Phishing (Best Project Award) (Live Demo)** [↗](#) | *Next.js, React.js, JavaScript, OpenAI API*

Jan – Apr 2025

- Led a 4-member team to build an **AI-powered** platform detecting phishing attacks and educating users in real time.
- Integrated **Azure OpenAI** for simulations, quizzes and a **virus-scanner** API, cutting false positives **27%**.
- Tested both API endpoints with **Postman** and helped isolate integration issues during simulation development.
- Added **Playwright end-to-end tests** for core threat-analysis, increasing coverage of key user scenarios by **30%**.

## Extracurricular Activities

**Langara Global Volunteer** | *Team Member & Social Media Ambassador*

Dec 2023 – Dec 2024

- Collaborated with **12** volunteers and **2** supervisors to organize **six** student engagement events per semester, including Explore Vancouver activities, helping international students adjust to campus life and build community.
- Managed event budgets and plan in **Word**, **Outlook**, **Excel** and **Powerpoint**, improving accountability in reporting

## Education

**Simon Fraser University** | **Transfer From Langara College**

Burnaby, BC

*Bachelor of Science in Computer Science*

*Jan 2023 - Jan 2028*